Addressing the global misinformation crisis one flag at a time

RedFlag

The future of fact checking

By Team Bronze
Old Dominion University
CS 410
Spring 2026

Table of Contents

Introduction
- 1: Redflag
- 2: Table of contents
- 3: Our team

The Problem
- 4: Social Media vs News
- 5: Misinformation in our Digital Ecosystem
- 6: Distrust is growing
- 7-8: Why it persists
- 9: Why does this matter?
- 10: Problem Characteristics

The Solution
- 11: The Solution
- 12: Solution Characteristics

Design
- 13: How will RedFlag work?
- 14: RedFlag Design Principles
- 15: Problem Process Flow
- 16: Solution Process Flow
- 17: Major Functional Components
- 18: Competition Matrix
- 19-21: Risk Matrices

Sources and extra information
- 22-23: References
- 24: Closing Statement
- 25: Glossary
- 26: Elevator Pitch

Our *team*

Ayden Garcia

TITLE: Team Lead

agarc046@odu.edu

James Moore

TITLE: Front-End Developer

jmoor072@odu.edu

Cameron Zell

TITLE: Backend Developer

czell002@odu.edu

Trey Leverett

TITLE: Web Developer

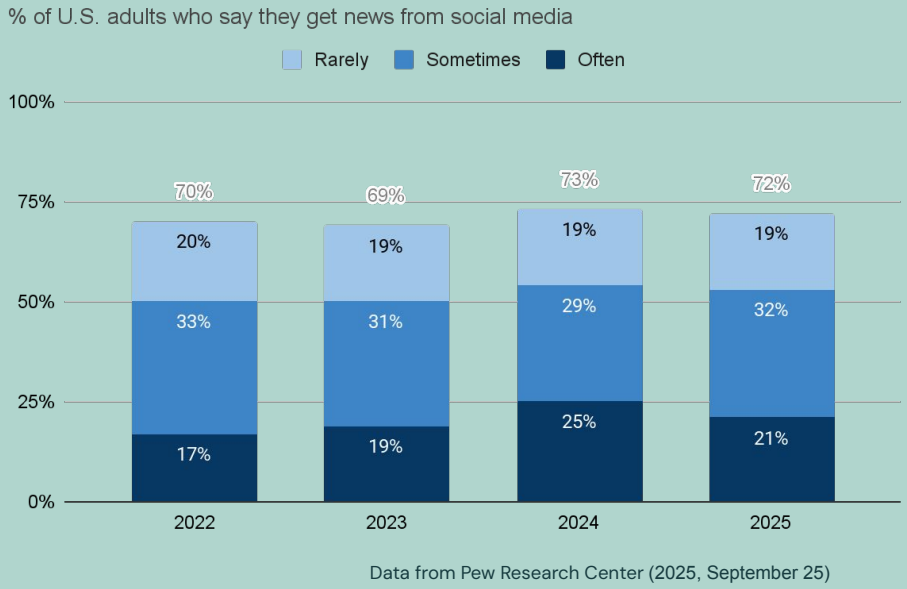
tleve002@odu.edu

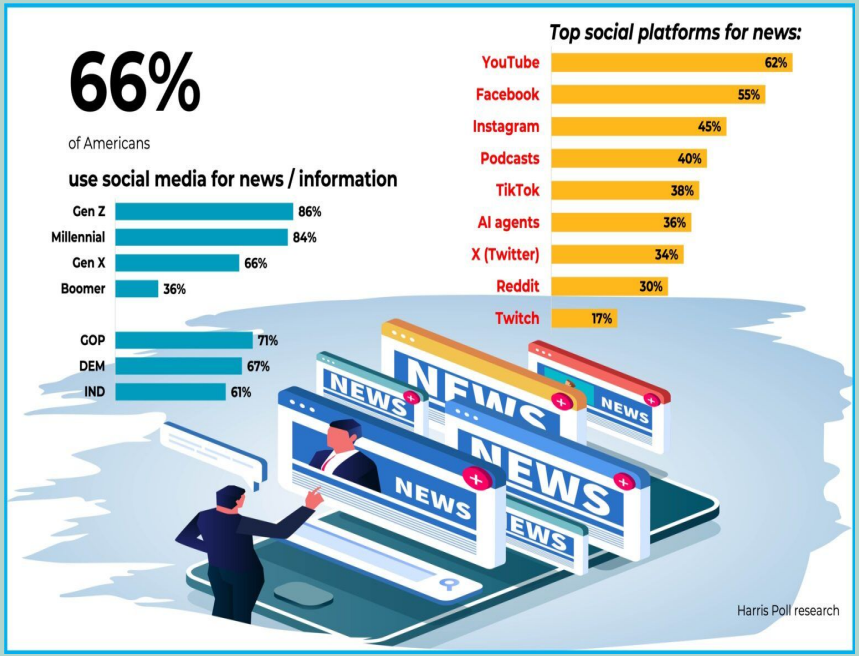
Riley Reynolds

TITLE: Graphic Designer

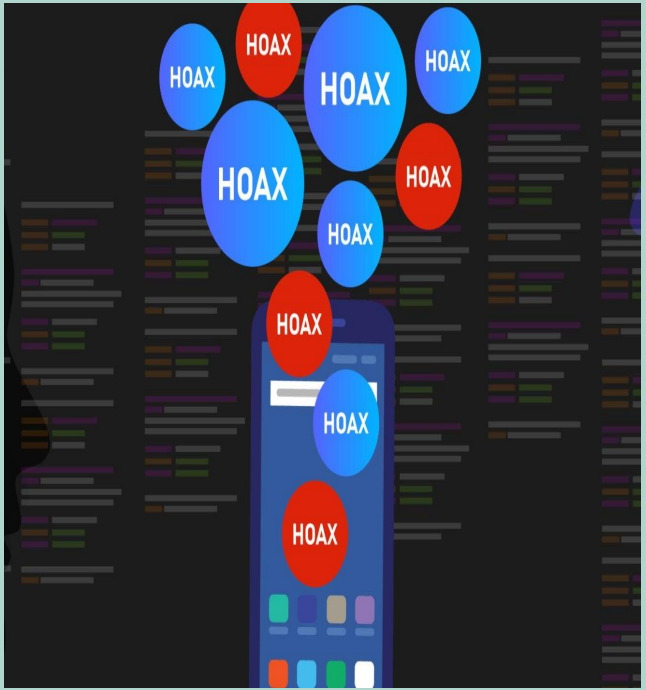
areyn017@odu.edu

Social Media as News



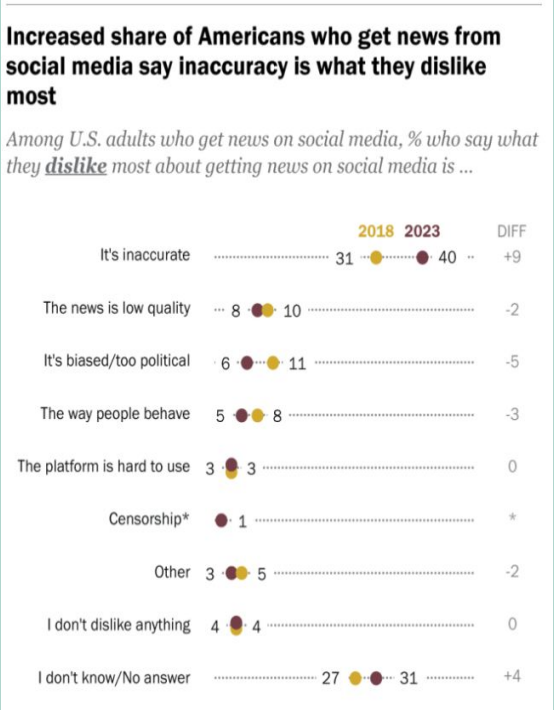


Misinformation in our Digital Ecosystem



- 94% of people report encountering misinformation online (The Alan Turing Institute, 2024)
- 38.2% of U.S. news consumers report unknowingly sharing fake news. (Redline Digital, 2024)

Distrust is growing



Pew Research Center. (2024, February 7)

- Concerns about inaccurate information on social media are rising (Pew Research Center, 2024)
- 40% of Americans who get news from social media cite inaccuracy as their biggest concern. (Pew Research Center, 2024)
- Only 37% say they trust information from social media platforms. (Pew Research Center, 2023)

So why does it persist?

- People don’t pay attention to what they’re consuming/sharing (Ordway, D 2021)
- Social media platforms reward engagement over accuracy (Ceylan, G 2023)

Evidence shows it is primarily a mindless action, done out of passivity and without care, and current systems only encourage it

- Community based fact-checking is too slow, corrections don't come in until long after posts have taken off (Blanding, M 2025)
- Professional human fact-checking is accurate but costly, and limited coverage makes real-time intervention impractical
- However, purely AI based fact checking often only compounds the problem as AI checkers have been found to be wrong 37-94% of the time (Ford 2025)
- Overconfident AI judgements can reinforce misinformation rather than correcting it
- Going to a separate website is time consuming and requires extra effort

Current Fact-checkers aren't effective

Why does this matter?

When misinformation spreads:

- Public trust declines (Pew Research Center, 2023)
- False information spreads unintentionally (Redline Digital, 2024), which causes countless negative outcomes including
  - Decline in public health (Denniss 2025)
  - Reduced trust in democratic institutions and increased polarization (Royal Holloway University of London 2024)
  - Hindering environmental efforts
  - Influencing the economy (Chen 2023)
  - And more

The fact checking tools of today are either too niche or bothersome to tackle this misinformation epidemic.

RedFlag

Problem Characteristics (THE GAP)

Lack of Universal Tool

Many existing verification tools are platform-specific and not unified. For example:

Community notes on X, NewsGuard for news sites, LLMs when manually provided context.

No Motivation to Verify

Platforms and algorithms encourage instant sharing with one click, while verifying claims may take minutes. Rationally, users choose not to spend this time and effort.

Lack of Credibility Awareness

Users receive no real-time credibility signals while browsing.

Inconvenience and Disconnected Verification

Fact-checking often requires manual searching outside the platform.

Algorithms Reinforce Bias

User's algorithms feed them content they inherently agree with which makes misinformation easier to spread.

RedFlag is a cross-platform fact checking (RedFlag) and bias indicator (BaisStar) running behind the scenes to tackle misinformation. When users are consuming the news any inaccuracies will be marked instantly with credible sources upon request, to prevent the unknowing spread of misinformation in today's information age. Furthermore on politically divisive issues RedFlag will star the content and offer different viewpoints from the left, right, and center at the users choosing.

RedFlag

Solution Characteristics (Closing THE GAP)

Convenience and Verification	Awareness of Credibility	Easily Accessible, Unprompted and Instant Verification	One Universal Tool	Sharing different points of view
Fact-checking is quick and painless, requiring little input from the user.	Users are actively informed of the credibility of sources they view while they're browsing.	Users are warned and informed before sharing potentially false claims, discouraging impulsive sharing.	A general tool for uncovering misinformation, no matter where the user browses.	Users are provided with a multitude of sources leaning in different areas of the political spectrum.

How will RedFlag work?

Integrated Detection

- Integrated by a chrome extension and or desktop application scanning the users screen to find misinformation at all times.

Real-Time Detection

- Highlights questionable information immediately as it is presented on screen.
- Also highlights political content to broaden users information scope and tackles social media algorithms.

Frictionless & Complete

- Highlights require no user input.
- Can deliver detailed overview on request.

What are RedFlag's Design Principles?

Unbiased political alignment

- RedFlag will not support any specific political standpoint
- No explicit ideological labeling of content
- Evidence based evaluation
- On disputed topics, RedFlag will use multiple sources and perspectives

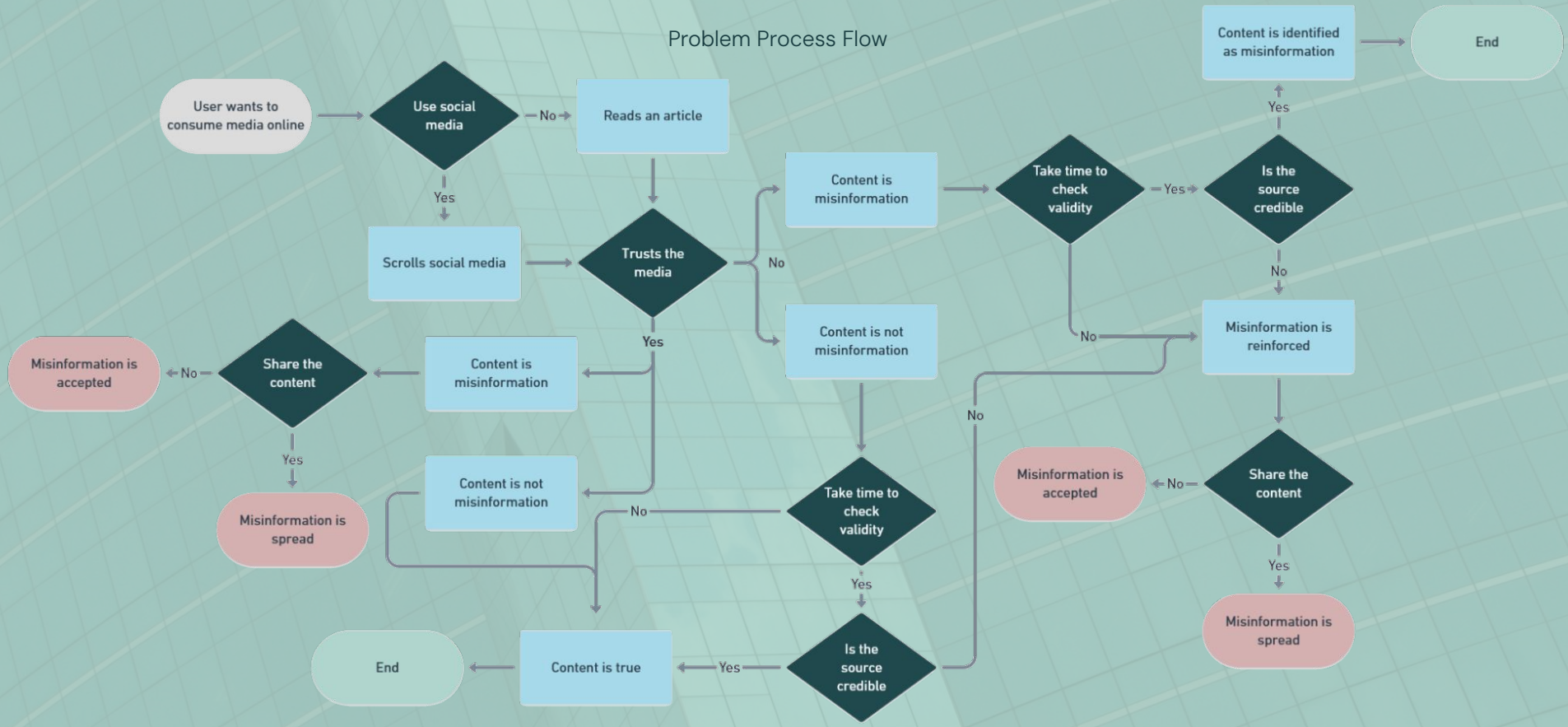
Uninterrupted flow

- Pop-ups would interrupt user's browsing experience
- Highlighting text is significantly less intrusive

Does NOT replace professional fact-checkers

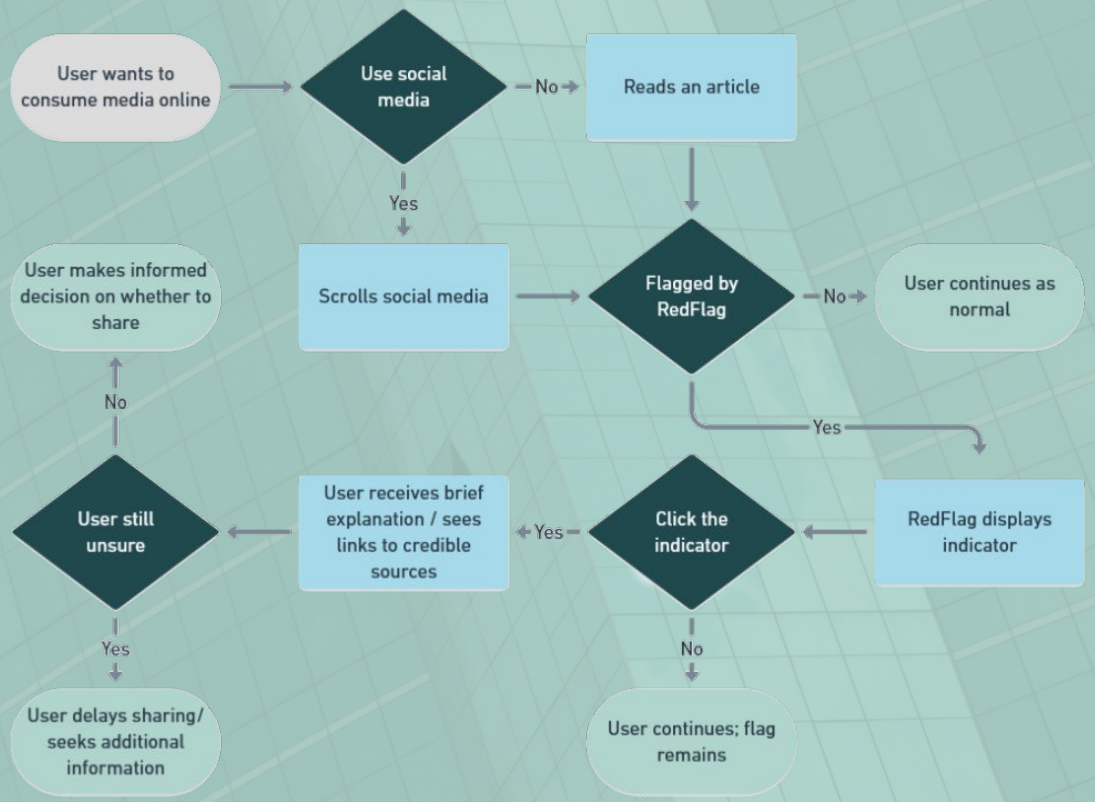
- Supports decision-making without dictating it
- Optional context is provided upon the user's request
- Supports journalism, but does not substitute it
- Does not act as a final authority on truth

Problem Process Flow



Made with Whimsical

Solution Process Flow



Major Functional Component Diagram – RedFlag

Presentation Layer

Media Consumers
- Social media users
- News readers

Devices
- Browser
- Mobile

RedFlag Extension
- Real-time alerts
- Balanced view panel

Application Layer

RedFlag Core Server

Modules:
- Content Detection
- Misinformation Analysis
- Fact Verification
- Balanced Perspectives Engine
- Alert & Recommendation Generator

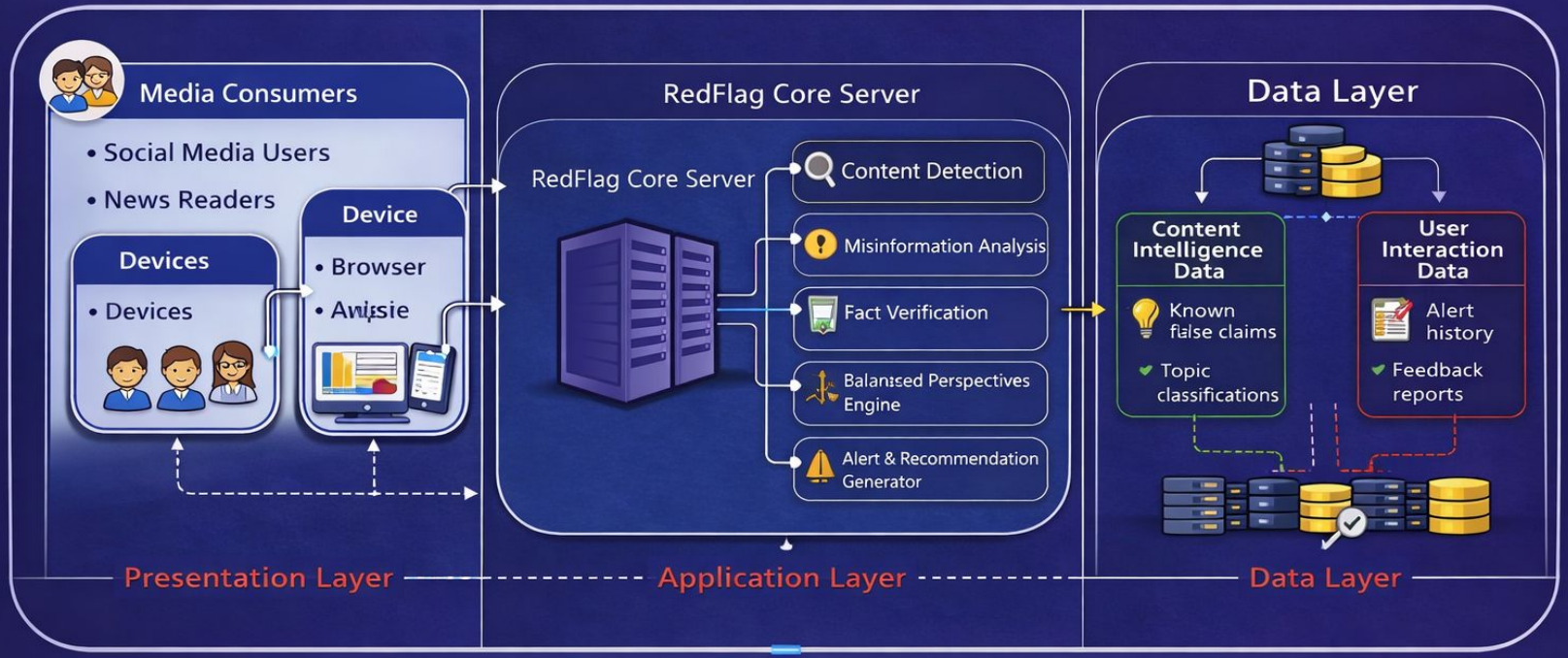
Data Layer

Content Intelligence Data
- Known false claims
- Topic classifications

Source Credibility Database
- Reliability scores
- Political alignment labels

User Interaction Data
- Alert history
- Feedback reports

Major Functional Component Diagram – RedFlag



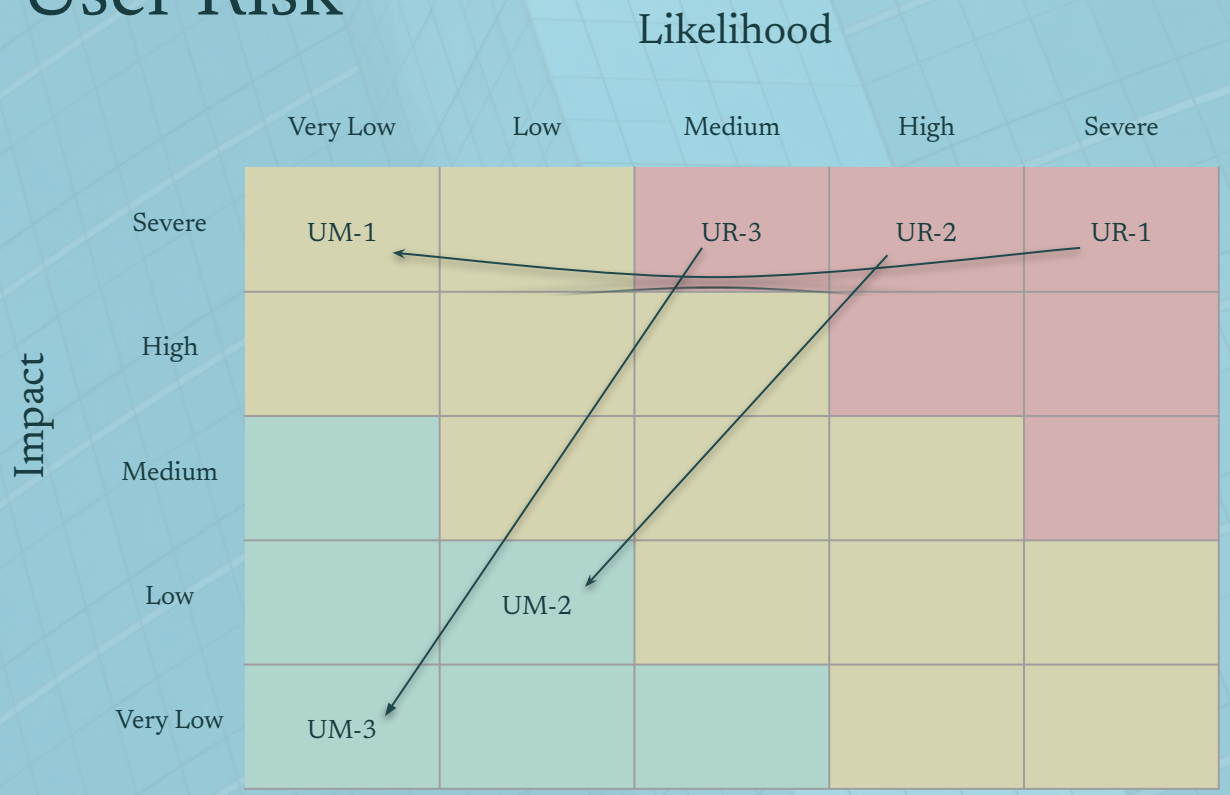
Competitors vs Redflag

	Fact checker databases (Example: Google Fact Check Explorer)	News Guard	Community Notes	AI/LLM
FEATURES	• Search for stories by keyword, person, or image that have been debunked • Fact check other stories yourself	• Rates trustworthiness of a news source on a scale from 1-100 • Team of journalists fact checking	• Allows users to add contextual explanations or fact checks to posts • Other contributors rate notes for helpfulness, only helpful notes are shown	• Ask AI the veracity of a claim/source (e.g. @Grok is this true?) • AI uses its dataset to respond
STRENGTHS	• Quickly able to search for rumors or claims being made online • Aggregate source of online disinformation	• Real human input on trustworthiness • Partnered with organizations for better fact checking	• Posts remain visible with extra context added • Uses a diverse group of users to reduce bias	• Ease of use- users already familiar with service, easily available, immediate response • Low manpower required
WEAKNESSES	• Allows biased sources to fact check other sources • Does not provide input on every claim made online • No verified source of claims	• Possible bias due to human fact checkers	• Slow response time makes notes appear long after a misleading post has spread • Organized groups can attempt to manipulate note ratings	• Frequently incorrect • Confidently wrong, fooling users • Dataset is fallible
	VS Fact checker databases	VS News Guard	VS Community Notes	VS AI
WHAT REDFLAG DOES BETTER / DIFFERENT	• Rather than search for stories, it will be real time when reading articles	• Works more actively to inform the user rather than passively on standby to check	• With the browser extension it will work for more than just online posts, and be faster due to a centralized database	• Will use AI solely increasing work flow of fact checking, but not act as a source of authority on any subject matter

Legend

Low Risk	Medium Risk	High Risk

User Risk



Risks

UR-1: Misinformation is not flagged or correct information is flagged.

UR-2: Users question credibility and stop using the tool.

UR-3: Intrusive UI or slow performance leads to user abandonment.

Mitigations

UM-1: Continuous AI retraining with updated misinformation datasets and multiple fact-check sources.

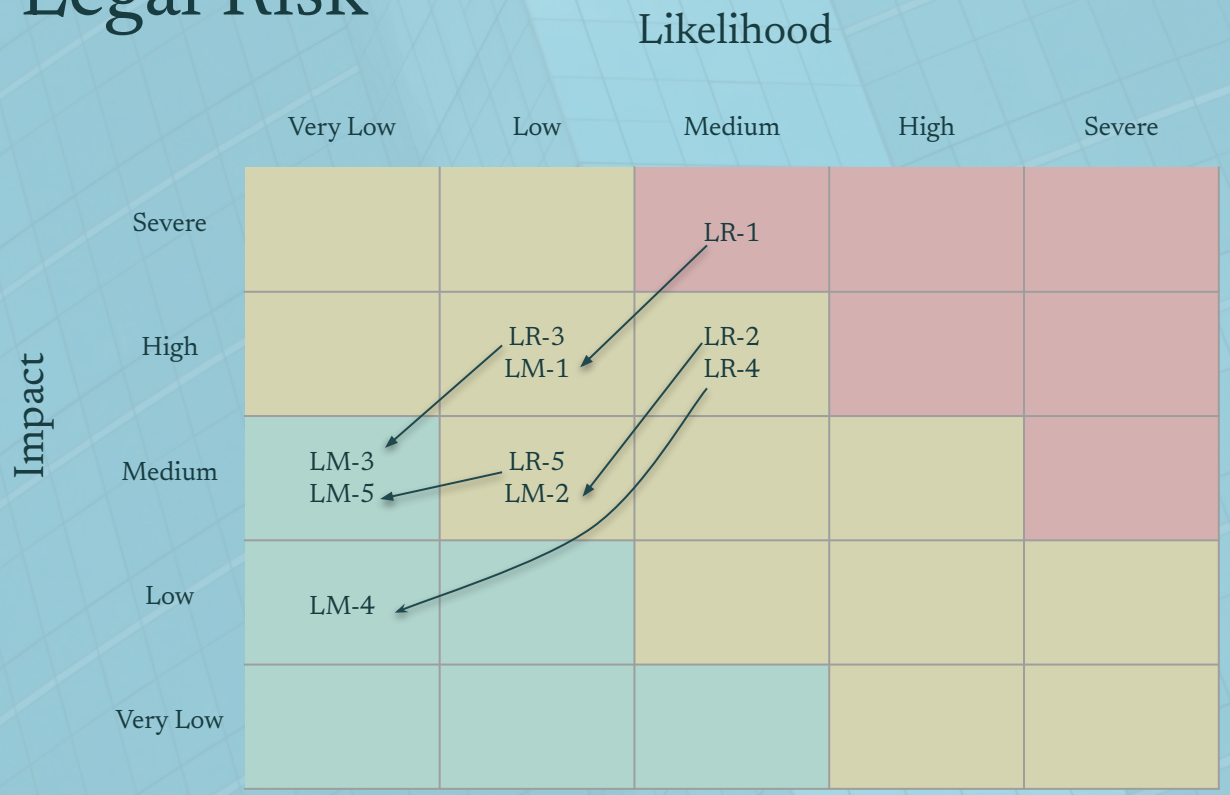
UM-2: “Why this was flagged” transparency feature with cited sources.

UM-3: Customizable alerts and performance optimization for smoother UX.

Legend

Low Risk	Medium Risk	High Risk

Legal Risk



Risks

LR-1: Improper handling of personal data
LR-2: Platform Terms of Service Violations resulting in removal from app stores
LR-3: Claims of reputational damage from incorrect flagging.
LR-4: Unauthorized use of protected datasets or APIs.
LR-5: Increased oversight of AI-based content moderation tools.

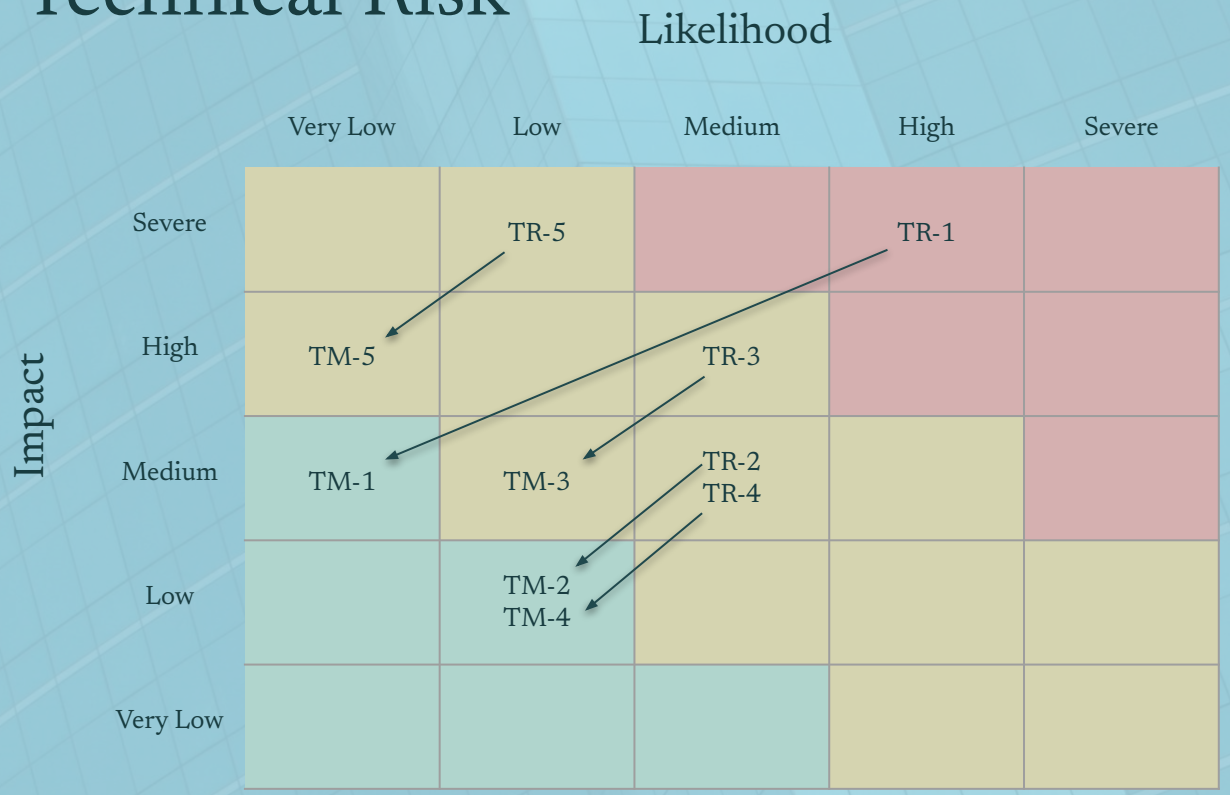
Mitigations

LM-1: Local data processing when possible and encrypted data transmission.
LM-2: Routine compliance reviews of browser store and API policies.
LM-3: Use of neutral phrasing such as "Potentially Misleading" instead of definitive claims.
LM-4: Use of licensed APIs and public domain datasets only.
LM-5: Clear disclaimer stating RedFlag is informational and not legal or professional judgment.LM-1: Local data processing when possible and encrypted data transmission.
LM-2: Routine compliance reviews of browser store and API policies.
LM-3: Use of neutral phrasing such as "Potentially Misleading" instead of definitive claims.
LM-4: Use of licensed APIs and public domain datasets only.
LM-5: Clear disclaimer stating RedFlag is informational and not legal or professional judgment.

Legend

Low Risk	Medium Risk	High Risk

Technical Risk



Risks

TR-1: AI model produces biased or inconsistent results.

TR-2: API usage costs exceed projected operational budget.

TR-3: Slow processing or latency reduces user adoption.

TR-4: Extension malfunctions across different browsers or software updates.

TR-5: Security vulnerabilities expose the system to potential exploitation.

Mitigations

TM-1: Routine benchmarking, bias testing, and model performance audits.

TM-2: Rate limiting, caching frequent queries, and cost monitoring tools.

TM-3: Asynchronous API calls and backend optimization.

TM-4: Cross-browser testing and staged rollout deployment.

TM-5: Code audits, encryption protocols, and regular penetration testing.

References

News Guard https://www.newsguardtech.com/

Rand Org https://www.rand.org/research/projects/truth-decay/fighting-disinformation/search.html

Pew Research Center. (2024, February 7). *Many Americans find value in getting news on social media, but concerns about inaccuracy have risen*. https://www.pewresearch.org/short-reads/2024/02/07/many-americans-find-value-in-getting-news-on-social-media-but-concerns-about-inaccuracy-have-risen/

Pew Research Center. (2023, October 30). *How Americans' trust in information from news organizations and social media sites has changed over time*. https://www.pewresearch.org/short-reads/2023/10/30/how-americans-trust-in-information-from-news-organizations-and-social-media-sites-has-changed-over-time/

The Alan Turing Institute. (2024). *More than 9 in 10 people report encountering misinformation online*. https://www.turing.ac.uk/news/more-90-uk-population-have-encountered-misinformation-online

Marshall, L. (2025, November 5). Older adults share more political misinformation. Here's why. University of Colorado Boulder. https://www.colorado.edu/today/2025/11/05/older-adults-share-more-political-misinformation-heres-why

Sijing Chen, Lu Xiao, Akit Kumar. (2023, April) *Spread of misinformation on social media: What contributes to it and how to combat it* https://www.sciencedirect.com/science/article/abs/pii/S0747563222004630

Royal Holloway University of London (2024) *Social Media, Misinformation and Harmful Algorithms: What We Know, and Don't Know, about Political Online Safety.* https://committees.parliament.uk/writtenevidence/132906/pdf/

References

Stagwell. (2025, October 15). What the data say: 85% of young people get their news from social media. https://www.stagwellglobal.com/what-the-data-say-85-of-young-people-get-their-news-from-social-media/

Redline Digital. (2024). *Fake news statistics & facts*. https://redline.digital/fake-news-statistics/

Pew Research Center. (2025, September 25) *Social Media and News Fact Sheet.* https://www.pewresearch.org/journalism/fact-sheet/social-media-and-news-fact-sheet/

Ordway, D. (2021, March 23) *Why do Americans share so much fake news? One big reason is they aren't paying attention, new research suggests.* Neiman Lab. https://www.media.mit.edu/articles/why-do-americans-share-so-much-fake-news-one-big-reason-is-they-aren-t-paying-attention-new-research-suggests/

Ceylan, G (2023, March 31) *How Social Media Rewards Misinformation.* Yale Insights https://insights.som.yale.edu/insights/how-social-media-rewards-misinformation

Blanding, M (2025, January 23) *The End of Fact-Checking Increases the Dangers of Social Media* https://now.tufts.edu/2025/01/23/end-fact-checking-increases-dangers-social-media

Ford, M and Eisele, I. (2025, May 17) *Fact check: How trustworthy are AI fact checks?* https://www.dw.com/en/fact-check-hey-grok-is-this-true-how-trustworthy-are-ai-fact-checks/a-72539345https://www.dw.com/en/fact-check-hey-grok-is-this-true-how-trustworthy-are-ai-fact-checks/a-72539345

Emily Denniss, Rebecca Lindberg. (2025 April) *Social media and the spread of misinformation: infectious and a threat to public health* https://academic.oup.com/heapro/article/40/2/daaf023/8100645

Thank You

Glossary

- **Friction** - something that causes an application to not be seamless while working.
- **L.L.M.** - A Large Language Model (also known as a ChatBot) such as ChatGPT, Gemini, or Claude.
- **Misinformation** - Provably false or inaccurate information
- **Disinformation** - False or inaccurate information that is shared with the intention to mislead others
- **Confirmation Bias** - The tendency to interpret new information and evidence as confirmation of one's existing beliefs or theories, often while ignoring or discounting opposing evidence
- **Credibility Score** - A score given to a claim that is found online rating the credibility of the claim, or the likelihood that the claim is correct.

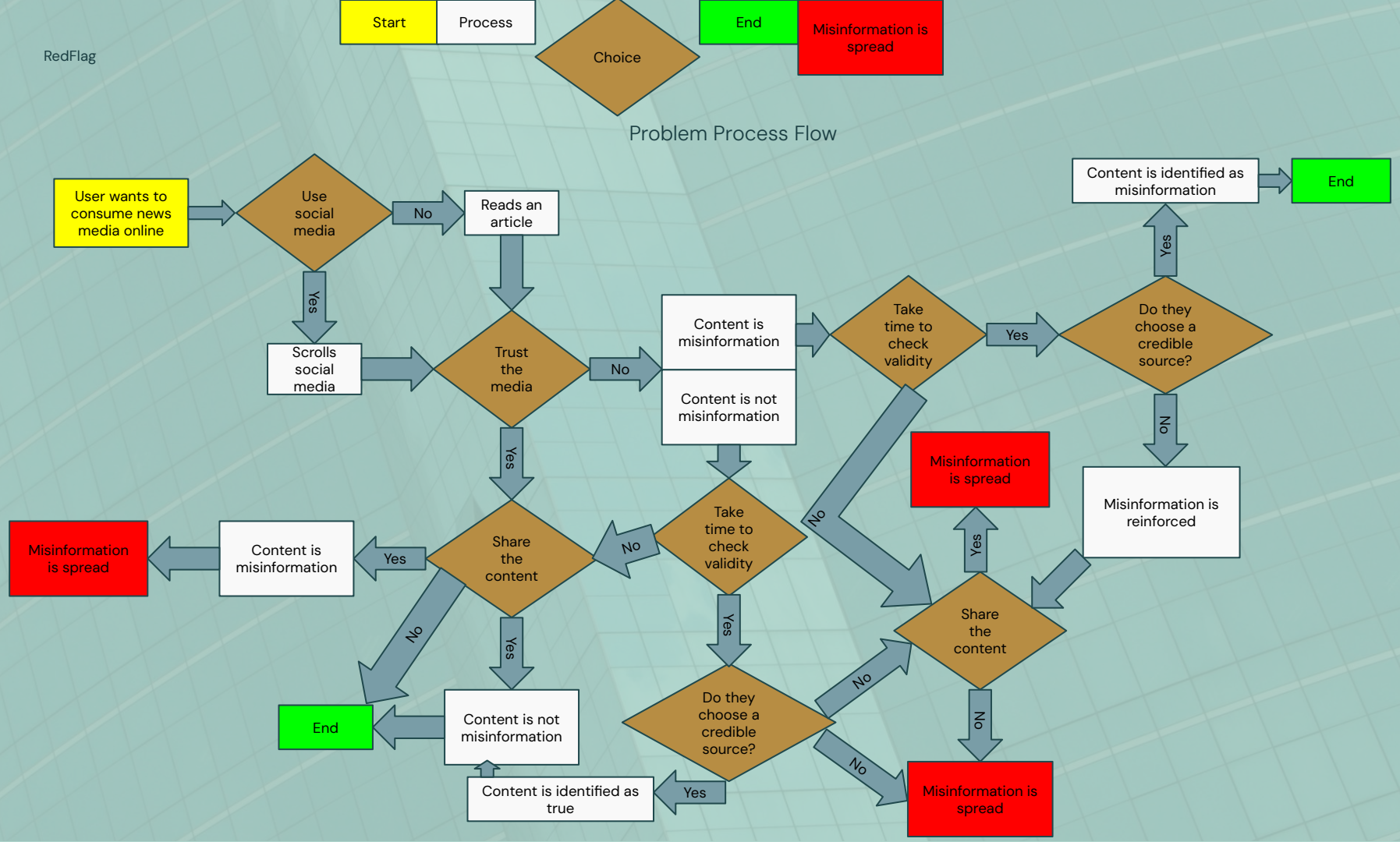
Elevator Pitch

RedFlag is a browser extension designed to reduce the spread of online misinformation. It runs in the background while users browse social media and news sites and automatically highlights potentially false or misleading content directly on the screen. Instead of relying on users to fact-check information themselves, RedFlag brings awareness to misinformation at the moment it's encountered. With one click, users can view a short explanation and links to credible sources, helping them make informed decisions before believing or sharing content.

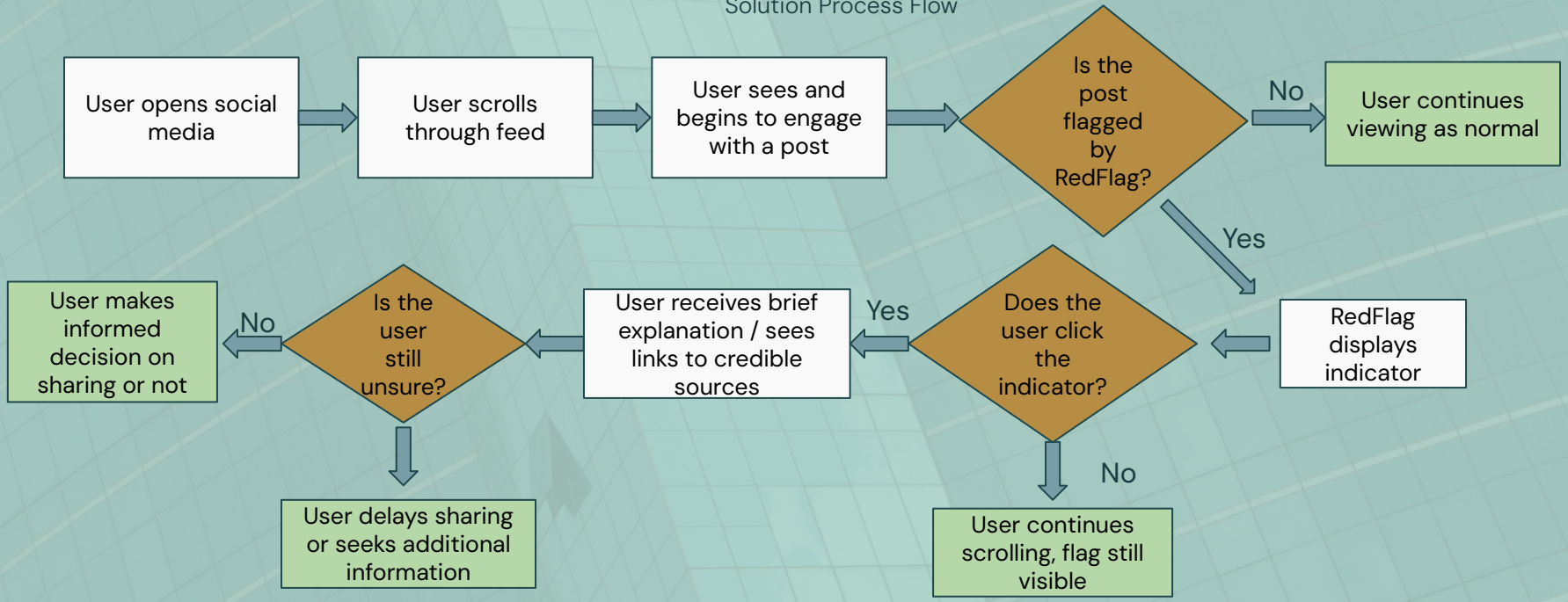
The problem plaguing online news

Despite the vastly popular concern in regarding digital inaccuracy, social media users frequently propagate misinformation because existing verification methods are inconvenient and require active user intervention. There is a critical need for a passive, real-time tool that bridges the gap between content consumption and fact-checking for daily social media users.

RedFlag


Start
Process
Choice
End
Misinformation is spread
Problem Process Flow
User wants to consume news media online
Use social media
No
Reads an article
Yes
Scrolls social media
Trust the media
No
Content is misinformation
Content is not misinformation
Take time to check validity
Yes
Do they choose a credible source?
Yes
Content is identified as misinformation
End
No
Misinformation is reinforced
Yes
Misinformation is spread
No
Share the content
No
Misinformation is spread
Yes
Take time to check validity
No
Share the content
Yes
Do they choose a credible source?
No
Misinformation is spread
Yes
Content is identified as true
Content is misinformation
Yes
Misinformation is spread
Share the content
No
End
Yes
Content is not misinformation

Solution Process Flow



Who does this affect?

- Users of social media
  - Teens
    - 85% of young people now get their news primarily from social media. (Stagwell, 2025)
  - 55+
    - On Twitter, 80% of fake news was shared by users over age 50 (Marshall, 2025)